SEVA SETU

Decentralized Cooperative Gig Platform with Fair-Wage Transparency & Priority Triage

Guiding Vision: 'Seva Se Samriddhi' (Prosperity Through Service) | **Governing Body:** Ministry of Cooperation, GoI | **Focus:** System Architecture, Scaling & 2-Minute Video Script

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EXECUTIVE PITCH HOOK (10 SECONDS):
Commercial gig aggregators extract 25% to 35% commission cuts from informal technicians while charging citizens opaque surge multipliers. **Seva Setu** organizes technicians and artisans under Primary Agricultural Credit Societies (PACS) and District Labour Federations, delivering **100% direct fair wages** with itemized cryptographic billing transparency, dual-OTP verification, and sub-15 minute emergency triage.

1. 2-Minute Video Presentation Framework (120-Second Pitch)

Designed for competitive hackathon jury evaluations (SIH 2026) and ministerial briefings. The 120 seconds are strictly partitioned across 5 clear, high-impact narrative blocks:

0:00 - 0:20 (20s) Hook & Crisis	0:20 - 0:45 (25s) Tri-Portal Model	0:45 - 1:15 (30s) Algorithmic Engines	1:15 - 1:40 (25s) National Scale	1:40 - 2:00 (20s) Impact & Closing

Time	Visual on Screen	Speaker Script / Teleprompter	Technical Highlights
0:00 – 0:20 (20 sec)	Split-screen showing private aggregator 25-35% cuts vs Seva Setu 100% fair wage guarantee badge.	"Commercial gig platforms charge citizens heavy markups while extracting up to 35% commission from informal workers. Introducing Seva Setu, a decentralized cooperative gig infrastructure operating under the Ministry of Cooperation's vision 'Seva Se Samriddhi'."	Platform Cooperatives Zero Aggregator Cut Fair-Wage Index
0:20 – 0:45 (25 sec)	Live UI tour showcasing 3 synchronized views: Citizen App, Worker Radar, Ministry Cockpit.	"Our architecture connects three primary actors: First, the Citizen Portal for itemized bookings and voice-first triage; Second, the Worker Radar with one-tap job acceptance and direct passbooks; and Third, the Ministry Cockpit tracking national wage dispersal and district skill deficits in real time."	Tri-Portal Ecosystem Leaflet Map Radar District Heatmaps
0:45 – 1:00 (15 sec)	Animated Transparent Invoice breakdown and Rapido-style radar broadcast wave.	"At the core, our Transparent Pricing Engine computes labor, toolkit, and Haversine transit costs with zero commission. Our Rapido-style Dispatch Engine matches nearest verified workers within a dynamic radius, auto-reassigning within 30 seconds if rejected."	Haversine Distance Dynamic Fare Matrix Fallback State Machine
1:00 – 1:15 (15 sec)	Dual-OTP arrival/completion flow and Gemini 1.5 Flash AI verification of ITI/Aadhaar credentials.	"Trust is cryptographic: Dual-OTP guarantees arrival and completion with photographic proof. Furthermore, our Gemini 1.5 Flash Multimodal OCR audits ITI trade certificates and Aadhaar credentials directly against cooperative registers."	Dual-OTP Escrow Gemini 1.5 Flash OCR Zero Ghost Billing
1:15 – 1:40 (25 sec)	Architecture schematic showing PostGIS, Uber H3 hex bins, Redis Streams, and Edge offline fallback.	"To scale nationally across 100,000+ PACS, we utilize PostGIS and Uber H3 hexagonal spatial indexing for sub-millisecond proximity queries. Redis Streams and distributed locks ensure zero double-booking, while an autonomous edge fallback guarantees 100% offline resilience."	Uber H3 (Res 7-8) PostGIS ST_DWithin Redis Redlock Edge PWA Engine
1:40 – 2:00 (20 sec)	Ministry telemetry showing INR 4.8M direct DBT wages, INR 1.8M citizen savings, and Government crest.	"By converting gig labor into cooperative equity, Seva Setu protects workers, reduces citizen costs, and builds a self-reliant rural-urban economic bridge. Seva Setu: Technology empowering grassroots cooperatives. Seva Se Samriddhi!"	DBT Settlement PACS Formalization Social Welfare Fund

2. System Architecture & Component Interactions

Seva Setu is architected as a 5-tier reactive platform designed for high availability, zero ghost billing, and verifiable institutional trust:

Tier	Technologies & Services	Core Responsibility & Guarantees

1. Presentation Layer	HTML5, Vanilla CSS Design System, Web Audio API, Leaflet.js, Web Speech API	Unified reactive interface hosting Citizen Portal, Worker Radar, and Ministry Cockpit with full English/Hindi localization.
2. Edge & Gateway	Netlify Edge / Cloudflare CDN, FastAPI, Pydantic, CORS Middleware	Global SSL termination, edge asset caching, rate limiting, and strict request validation across REST endpoints.
3. Domain Engine Core	Python 3.11 Microservices (pricing.py, dispatch.py, ai_ocr.py)	Autonomous pricing math, Rapido-style proximity matcher, fallback state machine, and Gemini 1.5 Flash multimodal OCR.
4. Spatial & Persistence	PostgreSQL 16 + PostGIS, Redis In-Memory Cache & Streams	Sub-millisecond spatial queries (ST_DWithin), ephemeral broadcast message queue, and atomic distributed booking locks.
5. GovTech Ecosystem	NPCI BharatPay UPI, UIDAI e-KYC, DigiLocker, National PACS Database	Direct Benefit Transfer (DBT) into District Central Cooperative Banks (DCCBs), certified trade credential verification.

3. Core Technical Engines & Mathematical Formulations

A. Transparent Fair-Wage Pricing Formulation

Commercial aggregators inflate final prices by 2.4x through hidden commission markups and dynamic surge multipliers. Seva Setu enforces a strictly itemized mathematical fare:

Mathematical Pricing Formulation:
- **Final Fare** = (Labor_base + Toolkit_cost + Transit_geo + Surcharge_urgency + Tax_GST + Cess_welfare) × (1 − Subsidy_NGO)
- **Geospatial Transit:** D_km = Haversine(Coords_citizen, Coords_worker); Transit_cost = max(INR 20.00, D_km × INR 8.00/km)
- **Cooperative Welfare Cess:** 3.5% of base labor → dedicated PACS worker healthcare and pension fund.
- **Aggregator Savings Delta:** Savings = (Final_Fare × 2.40) − Final_Fare (~58% citizen savings).

B. Rapido-Style Fallback State Machine with Dual-OTP

To prevent job abandonment and guarantee physical verification, the lifecycle transitions through an atomic state machine:

State	Trigger Event	System Action & Invariant	Fallback Behavior
1. REQUESTED	Citizen submits booking	Compute pricing; spatial query nearest candidates within 25km radius.	If 0 workers found → 404 radius error.
2. BROADCASTING	Primary worker alert dispatched	Worker phone vibrates + radar ping; 30-second decision timer starts.	If declined/timeout → auto-reassign to Worker #2.
3. ASSIGNED	Worker taps 'Accept'	Atomic lock created; worker details + START_OTP shared with customer.	If worker cancels → re-enter broadcast queue.
4. IN_PROGRESS	Technician arrives at premises	Technician inputs customer's START_OTP; work commences timer.	Zero ghost billing; work cannot start without OTP.
5. COMPLETED	Technician completes repair	Customer inputs END_OTP + photo proof upload; escrow released directly to DBT.	Immutable audit trail logged in Ministry database.

C. Gemini 1.5 Flash Multimodal Credential Verification

Technicians onboard by submitting Aadhaar cards and ITI/NSDC trade diplomas. The Gemini 1.5 Flash API parses these unstructured images into structured JSON verification objects, validating full names, document IDs, and trade specializations against government registries in under 2 seconds.

4. Enterprise Scalability & High-Load Architecture

To support national scaling across 100,000+ Primary Agricultural Credit Societies (PACS) and millions of concurrent citizens:

Architecture Domain	Scale Bottleneck	Seva Setu Enterprise Engineering Solution

Geospatial Proximity	Searching 500k+ worker GPS coordinates causes high CPU load.	**Uber H3 Hexagonal Grid (Res 7-8) + PostGIS:** Map coordinates to discrete hex IDs. Proximity searches use O(1) Redis set lookups (`kRing`) before spatial bounding-box refinement.
Dispatch Concurrency	Simultaneous requests for same technician cause race conditions.	**Redis Redlock & Distributed State:** Atomic lock acquisition (`SET lock:worker:101 booking_5001 NX EX 30`) ensures zero double-booking across distributed application nodes.
Real-Time Broadcast	HTTP polling degrades battery life and overwhelms REST APIs.	**WebSockets + Redis Pub/Sub:** Workers maintain lightweight persistent WebSocket channels. Job alerts and fallback timeouts are pushed instantaneously (<150ms).
Edge Offline Fallback	Rural connectivity loss disables cloud API operations.	**Dual-Mode Autonomous Edge:** The client contains a complete in-browser dispatch and pricing simulator that operates 100% offline, syncing state when reconnected.
Database Workload	Heavy analytics queries lock active booking tables.	**CQRS & Read Replicas:** Booking transactions write to Primary PostgreSQL; Ministry telemetry and heatmaps read from read replicas and Redis cache.

5. Production Database Schema (PostgreSQL + PostGIS DDL)

Structured relational schema enforcing referential integrity, geospatial indexing, and complete financial audit trails:

```
-- 1. Enable PostGIS Extension
CREATE EXTENSION IF NOT EXISTS postgis;

-- 2. Certified Cooperative Workers (Geospatial Indexing)
CREATE TABLE workers (
    id SERIAL PRIMARY KEY, coop_id INT REFERENCES cooperatives(id),
    name VARCHAR(120), phone VARCHAR(15) UNIQUE, skill VARCHAR(50),
    rating NUMERIC(2,1) DEFAULT 5.0, is_available BOOLEAN DEFAULT TRUE,
    geom GEOMETRY(Point, 4326), total_earned NUMERIC(12,2) DEFAULT 0.00
);
CREATE INDEX idx_workers_geom ON workers USING GIST(geom);

-- 3. Dual-OTP Escrow Bookings Lifecycle
CREATE TABLE bookings (
    id SERIAL PRIMARY KEY, customer_name VARCHAR(120), worker_id INT REFERENCES workers(id),
    service_type VARCHAR(50), plan VARCHAR(20), status VARCHAR(30) DEFAULT 'REQUESTED',
    start_otp VARCHAR(4), end_otp VARCHAR(4), base_labor NUMERIC(10,2), transit_cost NUMERIC(10,2),
    coop_welfare_cess NUMERIC(10,2), final_total NUMERIC(10,2), created_at TIMESTAMPTZ DEFAULT NOW()
);
```

6. Key Takeaways for Video Judges & Stakeholders

1. Socio-Economic Value: Replaces exploitative 30% private aggregator cuts with 100% direct fair wages deposited into worker bank accounts via DBT and District Central Cooperative Banks (DCCBs).
2. Complete Institutional Transparency: Itemized billing with transparent labor, toolkit, and Haversine transit costs, accompanied by a 3.5% cooperative welfare fund for health and pensions.
3. Cryptographic & AI Trust: Dual-OTP locks out ghost billing; Gemini 1.5 Flash multimodal OCR validates trade credentials against national registries.
4. High-Load Scalability: PostGIS and Uber H3 hexagonal spatial indexing paired with Redis Streams distributed state ensure nationwide scalability across 100,000+ PACS societies.